DIGITAL SAT & EST MATH COURSES

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EST I MATH • OFFICIAL EXAM • PRACTICE TEST

EST1 March 2021

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Algebra, functions, data analysis, probability and geometry from the official EST I paper. Work through all 58 questions under timed conditions, then check your answers with the answer key and the step-by-step solutions at the end.

58	MCQ	90 min
QUESTIONS	MULTIPLE CHOICE	TIME LIMIT

NAME _____ DATE _____ SCORE _____ /58

How to get the most from this practice test

- **Read** every question slowly and underline the key words.
- **Draw** a quick sketch for every geometry question before you solve it.
- **Solve** step by step and show your work.
- **Check** your answer by substituting it back before you choose.

RF

YOUR MATH COACH

Eng. Rabaa Farrag

Digital SAT & EST Math expert with **23 years of experience**, helping students master math and reach **700+**.

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Practice Questions

58 QUESTIONS

1

1. Isolate **D**.

A. $D = 2\sqrt{\frac{\pi I}{P}}$

B. $D = \frac{1}{2}\sqrt{\frac{P}{\pi I}}$

C. $D = (\frac{P}{4\pi I})^2$

D. $D = \sqrt{\frac{P}{2\pi I}}$

2

If the power of the source doubles (P'=2P), and the distance between the source and point A also doubles (D'=2D), the sound intensity becomes I'. What is the relationship between I' and I?

(A) I' = I
(B) I'= 4 I
(C) I' = I / 2
(D) I' = I / 4

3

The weekly salary S of Samantha is given by $S = 7hd + 75$ where h is the number of hours she works every day and d is the number of working days each week. If Samantha is to be promoted with a salary increase, which of the following terms must change?

(A) 7
(B) h
(C) d
(D) None of the above

4

In the xy-plane, the equation of the line (d) passing through A(-1,4) and perpendicular to line $3x - 2y = 5$ is:

(A) $y = \frac{3}{2}x + \frac{11}{2}$
(B) $y = -\frac{3}{2}x + \frac{5}{2}$
(C) $y = -\frac{2}{3}x + \frac{14}{3}$
(D) $y = -\frac{2}{3}x + \frac{10}{3}$

5

A researcher selected a random sample of 25 different brands of bottled water and measured the corresponding PH. He found out that the mean PH of the sample was 7.3 with an associated margin of error of 0.25. Which of the following is the best interpretation of the researcher's findings?

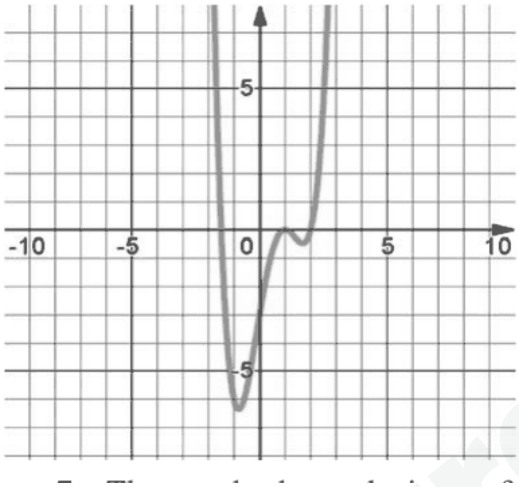
(A) Most water bottles in the market have a PH between 7.05 and 7.55
(B) All water bottles in the market have a PH between 7.05 and 7.55
(C) Any PH between 7.05 and 7.55 is a plausible value for the mean PH of the water bottles in the sample
(D) Any PH between 7.05 and 7.55 is a plausible value for the mean PH of the water bottles in

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6 If $h(x) = -x^2 + 3x - 2$ and $k(x) = -2x - 5$, what is the value of $h(k(-2))$?

(A) -6
(B) -4
(C) 0
(D) 2

7 The graph above depicts a function f(x). How many solutions does the equation f(x) = 0 admit?

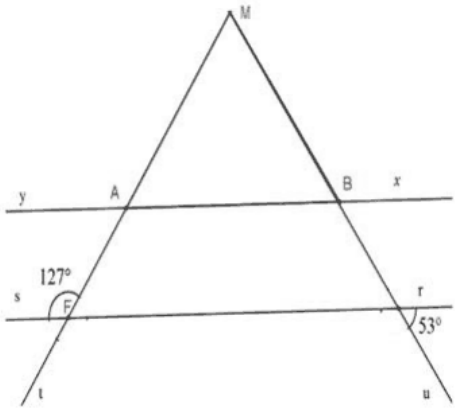


(A) 1
(B) 2
(C) 3
(D) 4

8 Which of the following angles have the same value of $\sin 32°$?

(A) $\sin(-32°)$
(B) $\sin 58°$
(C) $\cos(-32°)$
(D) $\cos 58°$

9 Knowing that lines (xy) and (rs) are parallel, what is the type of triangle ABM? (Figure is not drawn to scale)



(A) Scalene
(B) Isosceles
(C) Equilateral
(D) Right isosceles

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10 A rental car company charges its clients by the number of days d the car is rented. If the bill b of a client is given by the equation $b = 5d + 20$, what is the best interpretation of the term 5?

(A) The rental cost of the car per day
(B) The fixed cost of renting a car, no matter how many days it is rented
(C) The average number of days the car is rented
(D) The fees of renting a car for 7 days

11 The table below summarizes the grades out of 10 obtained by 4 players in a competition over a total of 5 rounds.

	Diego	Mark	Tyler	Toni
Round 1	7	3	8	10
Round 2	5	10	7	4
Round 3	8	9	6	9
Round 4	9	5	7	10
Round 5	4	8	8	7
Standard deviation	1.85	2.61	0.75	2.28

11. The winner is the player with the highest average. Who won the competition?

A. Diego
B. Toni
C. Tyler
D. Mark

12 Which player was the most consistent in terms of his grades?

	Diego	Mark	Tyler	Toni
Round 1	7	3	8	10
Round 2	5	10	7	4
Round 3	8	9	6	9
Round 4	9	5	7	10
Round 5	4	8	8	7
Standard deviation	1.85	2.61	0.75	2.28

(A) Mark
(B) Diego
(C) Toni
(D) Tyler

13 What is the resulting coefficient of x when -2x+3 is multiplied by -3x-2?

(A) -9
(B) -5
(C) 5
(D) 6

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14

$$\frac{-6x^2 + 5x + 2}{2x + 1}$$

14. Which of the following is equivalent to the above expression?

A. $-3x + 4 - \frac{2}{2x+1}$
B. $-3x + 4 + \frac{2}{2x+1}$
C. $-3x - \frac{2}{2x+1}$
D. $-3x + 4$

15 If m and k are positive numbers, which of the following expressions is equivalent to $(16k^{12}m^4)^{\frac{1}{4}}$?

(A) $4k^3m$
(B) $2k^3$
(C) $4k^3m^2$
(D) $2k^3m$

16 If

$$2^{x+1} = 8^{y-1}$$

and

$$3^{x-1} = 9^{y+1}$$

, what is the value of

$$x + y$$

?

17 If $h(x) = -x^2 + 3x - 2$ and $k(x) = -2x - 5$, what is the value of $h(k(-2))$?

18 If $|-2b - 3| \leq 7$, how many possible integer values of b are there?

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19 The table above shows several points from the graph of quadratic function q(x). What is q(-9)?

x	q(x)
-5	-7
-3	-15
-1	-15
1	-7
3	9
5	33

20 If $h(x) = -x^2 + 3x - 2$ and $k(x) = -2x - 5$, what is the value of $h(k(-2))$?

21 If

$$2^{x+1} = 8^{y-1}$$

and

$$3^{x-1} = 9^{y+1}$$

, what is the value of

$$x + y$$

?

(A) 14
(B) 17
(C) 21
(D) 24

22 **2.** If $-\frac{4}{5}x + 3 \geq 2 - \frac{1}{5}x$, what is the highest value of $\frac{3}{2}x + 4$?

A. 3.5
B. 4.5
C. 5.5
D. 6.5

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25 Using the same table, which measure of center is unchanged in both cities?

(A) Mode
(B) Mean
(C) Median
(D) None of the above

26 If $f(x) = -3 - 2x$ and $g(x) = \frac{-x^2}{6}$, which of the following could not be in the range of $f(g(x))$?

(A) -4
(B) -2
(C) 0
(D) 2

27 A shop sells two sizes of doughnuts: mini and regular. Mini doughnuts have a diameter of 1.8 inches while regular ones have a diameter of 3 inches. Approximately, by what percentage is a regular normal doughnut larger (in area) than a mini doughnut?

(A) 67%
(B) 67\pi%
(C) 78%
(D) 178%

28 Line (A) passes through the points $(-\frac{5}{2}; 0)$ and $(0; -\frac{2}{3})$. Which of the following lines will never intersect with line (A)?

(A) $y = -\frac{5}{2}x - 1$
(B) $y = \frac{5}{2}x - 1$
(C) $y = -\frac{2}{5}x - 1$
(D) $y = \frac{2}{5}x - 1$

29 If $(3^9)^{3^{12}} = 3^{3^x}$, what is the value of x?

(A) 9
(B) 14
(C) 21
(D) 27

30 An oven costs $150 less than 4 times the cost of a microwave. If the oven and the microwave cost together 725, how much more does the oven cost than the microwave?

(A) $175
(B) $275
(C) $375
(D) $550

31 From 2020 to 2021, the amount in Tom's bank account decreased by 11% to $49840. What was the initial amount in his bank account?

(A) $54820
(B) $44358
(C) $56000
(D) $45309

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32 If $(3^9)^{3^{12}} = 3^{3^x}$, what is the value of x?

(A) 9
(B) 14
(C) 21
(D) 27

33 **Questions 13 and 14 refer to the following information.**

The table below summarizes the results of a survey about the favorite school subject for a group of 350 students according to their educational stage.

	Primary	Secondary	Tertiary	To
Math	65	80	30	17
English	35	55	10	10
Science	15	10	25	5
History	10	5	10	2
Total	125	150	75	35

13. According to the data, which subject had the lowest percentage of likers among primary students?

A. Math
B. English
C. Science
D. History

34 If a secondary student is chosen randomly, what is the probability that he favors Math or Science?

(A) $\frac{9}{35}$
(B) $\frac{18}{25}$
(C) $\frac{8}{15}$
(D) $\frac{3}{5}$

35 Given a cone whose height is the quadruple of its diameter. If the volume of the cone is 72π, what is the diameter of its base?

(A) 4
(B) 5
(C) 6
(D) 9

36 If $11x - 7y = 6$ and $17x - 13y = 9$, what is the value of $x - y$?

(A) 0
(B) $-\frac{1}{2}$
(C) $\frac{1}{2}$
(D) -2

37 Anton has $\$20$ to spend on stationary. Pens (p) cost $\$1.4$ each, coloring crayons (c) are priced at $\$3.5$ per pack and highlighters (h) sell for $\$2$ each. He must buy a notebook for $\$5.5$ as well. Which of the following describes how many highlighters Anton can buy?

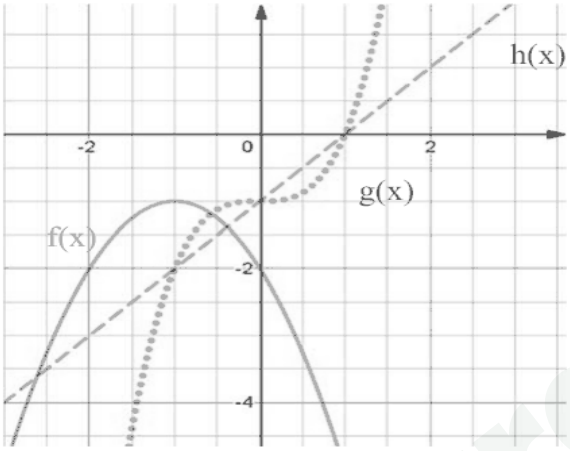
(A) $h \leq \frac{14.5-1.4p-3.5c}{2}$
(B) $h \leq \frac{14.5+1.4p+3.5c}{2}$
(C) $h \leq \frac{20-1.4p-3.5c}{2} - 5.5$
(D) $h \leq \frac{20-1.4p-3.5c}{2}$

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39 The general elections in a country provided a parliament composed of 35% deputies from the democratic party, 20% for the conservative party, 30% for the republican party and 27 members of the labor party. How many more democrats are there than conservatives?

(A) 9
(B) 18
(C) 27
(D) 36

40 Based on the graph above, what is the value of $f(-1) + g(-1) + h(-1)$?



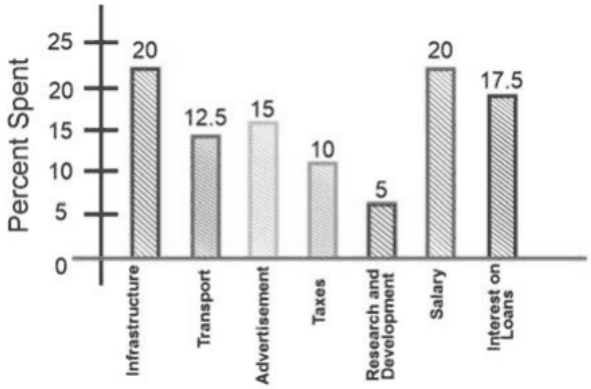
20. Based on the graph above, what is the

(A) -4
(B) -5
(C) -6
(D) -7

41 A painter paints walls at an average rate of 1.2 walls per hour. He is asked to paint a villa containing 18 rooms each containing 5.4 walls on average. How long will he be actively painting the villa?

(A) 2 days, 18 hours
(B) 3 days
(C) 3 days, 9 hours
(D) 3 days, 18 hours

42 The bar graph represents the percentage of money spent by a company on different sectors last year. If 200,000 dollars were spent on the transport sector, what was the budget used for salaries?



(A) $320,000
(B) $340,000

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47 Which of the following is equivalent to the quotient of the division given above: $\frac{\sqrt[4]{3}}{\sqrt[8]{3}}$?

(A) $\sqrt{3}$
(B) $\sqrt[4]{3}$
(C) $\sqrt[8]{3}$
(D) 9

48 The number of bees in a beehive rose from 2400 to 2520 in a month. How many bees do we expect to have in the next month knowing that the population experiences the same percent increase every month?

(A) 2600
(B) 2640
(C) 2646
(D) 2656

49 Which value of x could make the inequality $\frac{2}{3}(x+2)-x>5$ true?

(A) -12
(B) -10
(C) -8
(D) -6

50 Harvey wants to buy a new apartment. He estimates that he will need its area to be around 1400 square feet. But, the provider measures its units in square meters. If 1 meter is nearly 3.28 feet, how many square meters approximately will Harvey need?

(A) 427
(B) 130
(C) 4592
(D) 15062

51 The mean of x,y,z and t is 11.5. The mean of x,y,z,t and q is 11.8. What is the value of q?

52 The function g is defined by $g(x)=3x^2+kx-8$ and $g(-2)=-4$. What is the value of $g(-3)$?

53 What is the remainder of the division of $P(x)=4x^3-x^2-8x+6$ by $x-1$?

54 The concentration of solution in sugar is directly proportional to the mass of sugar dissolved in the solution. If the concentration of a solution is 1.5mol/L when 54g are dissolved, how many grams are dissolved if the concentration is 2.1mol/L?

55 If $h(x)=-x^2+3x-2$ and $k(x)=-2x-5$, what is the value of $h(k(-2))$?

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56 The table shows several values for the functions p(x) and q(x). What is the value of p(q(0))?

x	-3	-2	-1	0	1	2
p(x)	7	4	1	-2	-5	-8
q(x)	-7	-5	-3	-1	1	3

57 If $h(x) = -x^2 + 3x - 2$ and $k(x) = -2x - 5$, what is the value of $h(k(-2))$?

58 Trevor has $600 to spend on shopping, 25 percent of which he spent to buy groceries. If he budgets 30 percent of the remaining amount f
120 for cooking utensils and spends the rest on gaming accessories, what percentage of the original $600 did he spend on gaming accessories? (Input your answer without the % sign)

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Answer Key

58 ANSWERS

Q	ANSWER	Q	ANSWER
1	B	30	C
2	C	31	C
3	A	32	B
4	D	33	D
5	D	34	D
6	A	35	C
7	D	36	C
8	D	37	A
9	B	38	C
10	A	39	C
11	B	40	B
12	D	41	C
13	B	42	A
14	A	43	C
15	D	44	A
16	0.2	45	A
17	60	46	C
18	8	47	C
19	33	48	C
20	14	49	A
21	C	50	B
22	D	51	13
23	B	52	7
24	B	53	1
25	D	54	75.6
26	A	55	0.28

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Detailed Solutions

58 SOLUTIONS

1 **Answer: B**

See the answer key.

2 **Answer: C** I′ = I / 2

$I' = \frac{2P}{4\pi(2D)^2} = \frac{2P}{4\pi(4D^2)} = \frac{2}{4}(\frac{P}{4\pi D^2}) = \frac{1}{2}I.$

3 **Answer: A** 7

A promotion implies an increase in her rate of pay. 'h' and 'd' are the hours and days she works, while '75' is the fixed base. So the hourly rate '7' must change.

4 **Answer: D** $y = -\frac{2}{3}x + \frac{10}{3}$

Line $3x - 2y = 5 \Rightarrow y = \frac{3}{2}x - \frac{5}{2}$. Slope is $3/2$. Perpendicular line slope is $-2/3$. Using point-slope: $y - 4 = -\frac{2}{3}(x + 1) \Rightarrow y = -\frac{2}{3}x - \frac{2}{3} + 4 = -\frac{2}{3}x + \frac{10}{3}$.

5 **Answer: D** Any PH between 7.05 and 7.55 is a plausible value for the mean PH of the water bottles in the market

The confidence interval provides a range of plausible values for the population parameter (the true mean of ALL water bottles in the market), not just the sample or individual bottles.

6 **Answer: A** -6

$k(-2) = -2(-2) - 5 = 4 - 5 = -1$. Then $h(-1) = -(-1)^2 + 3(-1) - 2 = -1 - 3 - 2 = -6$.

7 **Answer: D** 4

By observing the provided graph, the curve intersects the horizontal x-axis at 4 distinct points. Therefore, there are 4 solutions to f(x) = 0.

8 **Answer: D** $\cos 58°$

Sine and cosine are co-functions: $\sin(\theta) = \cos(90° - \theta)$. Thus, $\sin(32°) = \cos(90° - 32°) = \cos(58°)$.

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9 **Answer: B** Isosceles

Let the angles inside ΔABM at A be α and at B be β. $\alpha = 180^\circ - 127^\circ = 53^\circ$. Because lines are parallel, alternating interior angles are equal, meaning the angle forming the 53° translates directly, so $\beta = 53^\circ$. Since $\alpha = \beta = 53^\circ$, the triangle is isosceles.

10 **Answer: A** The rental cost of the car per day

In the linear equation $b = 5d + 20$, '5' is the slope, which represents the rate of change of the bill with respect to the number of days. That is the cost per day.

11 **Answer: B**

See the answer key.

12 **Answer: D** Tyler

Consistency is measured by standard deviation. The lower the standard deviation, the more consistent the scores. Tyler has the lowest standard deviation (0.75).

13 **Answer: B** -5

$(-2x + 3)(-3x - 2) = (-2x)(-3x) + (-2x)(-2) + (3)(-3x) + (3)(-2) = 6x^2 + 4x - 9x - 6 = 6x^2 - 5x - 6$. The coefficient of x is -5.

14 **Answer: A**

See the answer key.

15 **Answer: D** $2k^3m$

Apply the exponent to each term: $16^{1/4} = 2$. $(k^{12})^{1/4} = k^3$. $(m^4)^{1/4} = m^1$. Result is $2k^3m$.

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16 **Answer: 0.2**

Express both sides of each equation with the same base.
First equation: $2^{x+1} = (2^3)^{y-1} \Rightarrow 2^{x+1} = 2^{3y-3}$.
So, $x + 1 = 3y - 3 \Rightarrow x - 3y = -4$.
Second equation: $3^{x-1} = (3^2)^{y+1} \Rightarrow 3^{x-1} = 3^{2y+2}$.
So, $x - 1 = 2y + 2 \Rightarrow x - 2y = 3$.
Subtract the first equation from the second:

$$(x - 2y) - (x - 3y) = 3 - (-4)$$

$$y = 7$$

Substitute $y = 7$ into $x - 2y = 3$:

$$x - 2(7) = 3 \Rightarrow x - 14 = 3 \Rightarrow x = 17$$

The value of $x + y = 17 + 7 = 24$.

17 **Answer: 60**

$k(-2) = -2(-2) - 5 = 4 - 5 = -1$. Then $h(-1) = -(-1)^2 + 3(-1) - 2 = -1 - 3 - 2 = -6$.

18 **Answer: 8**

$-7 \leq -2b - 3 \leq 7 \Rightarrow -4 \leq -2b \leq 10 \Rightarrow -5 \leq b \leq 2$. The integers are -5, -4, -3, -2, -1, 0, 1, 2. There are 8 total values.

19 **Answer: 33**

Because $q(-3) = q(-1) = -15$, the axis of symmetry is exactly in the middle of -3 and -1, yielding $x = -2$. The distance from the axis of symmetry to $x = -9$ is $|-9 - (-2)| = 7$. The point $x = 5$ is also distance 7 from the axis of symmetry ($|5 - (-2)| = 7$). Because quadratics are perfectly symmetric, $q(-9) = q(5) = 33$.

20 **Answer: 14**

$k(-2) = -2(-2) - 5 = 4 - 5 = -1$. Then $h(-1) = -(-1)^2 + 3(-1) - 2 = -1 - 3 - 2 = -6$.

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21 **Answer: C** 21

Express both sides of each equation with the same base.
First equation: $2^{x+1} = (2^3)^{y-1} \Rightarrow 2^{x+1} = 2^{3y-3}$.
So, $x + 1 = 3y - 3 \Rightarrow x - 3y = -4$.
Second equation: $3^{x-1} = (3^2)^{y+1} \Rightarrow 3^{x-1} = 3^{2y+2}$.
So, $x - 1 = 2y + 2 \Rightarrow x - 2y = 3$.
Subtract the first equation from the second:

$$(x - 2y) - (x - 3y) = 3 - (-4)$$

$$y = 7$$

Substitute $y = 7$ into $x - 2y = 3$:

$$x - 2(7) = 3 \Rightarrow x - 14 = 3 \Rightarrow x = 17$$

The value of $x + y = 17 + 7 = 24$.

22 **Answer: D**

See the answer key.

23 **Answer: B** b > 0

The parabola opens upwards, so $a > 0$. It intersects the y-axis (where $x = 0$) at $y = 0$, indicating $c = 0$. The vertex's x-coordinate is positive ($x = -b/2a > 0$). Since a is positive, b must be negative ($b < 0$). Thus 'b > 0' is the false statement.

24 **Answer: B**

See the answer key.

25 **Answer: D** None of the above

For City A: Total=80, Median is 40th/41st value (which falls in 2 cars). Mode = 2. Mean = $171/80 = 2.1375$. For City B: Total=80, Median is 40th/41st value (which falls in 3 cars). Mode = 3. Mean = $205/80 = 2.5625$. Thus, Mode, Median, and Mean all change between the two cities.

26 **Answer: A** -4

$f(g(x)) = f(\frac{-x^2}{6}) = -3 - 2(\frac{-x^2}{6}) = -3 + \frac{x^2}{3}$. Because $\frac{x^2}{3} \geq 0$, the minimum value of $f(g(x))$ is -3. Thus, any value less than -3 (such as -4) cannot be in the range.

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27 **Answer: D** 178%

The ratio of their diameters is $3/1.8 = 1.666... = 5/3$. The ratio of their areas is $(5/3)^2 = 25/9 \approx 2.777...$. This means the area is $\approx 277.8\%$ of the mini doughnut's area. The percentage by which it is *larger* is $(2.777 - 1) \times 100\% \approx 178\%$.

28 **Answer: C** $y = -\frac{2}{5}x - 1$

Two lines will never intersect if they are parallel (i.e. strictly have the same slope). Slope of Line A is $m = -\frac{2}{5}$. The option that matches is $y = -\frac{2}{5}x - 1$.

29 **Answer: C** 21

$(3^9)^{3^{12}} = 3^{9 \times 3^{12}} = 3^{3^2 \times 3^{12}} = 3^{3^{14}}$. Therefore, $3^x = 3^{14} \Rightarrow x = 14$.

30 **Answer: C** $375

Let M be the microwave cost. Oven $O = 4M - 150$. Together: $O + M = 725 \Rightarrow (4M - 150) + M = 725 \Rightarrow 5M = 875 \Rightarrow M = 175$. Microwave = $175, Oven = 550$. Difference: $550 - 175 = 375$.

31 **Answer: C** $56000

Let the initial amount be A. A decrease by 11% leaves 89% of the initial amount.
$0.89A = 49840$.
$A = \frac{49840}{0.89} = 56000$.

32 **Answer: B** 14

$(3^9)^{3^{12}} = 3^{9 \times 3^{12}} = 3^{3^2 \times 3^{12}} = 3^{3^{14}}$. Therefore, $3^x = 3^{14} \Rightarrow x = 14$.

33 **Answer: D**

See the answer key.

34 **Answer: D** $\frac{3}{5}$

The condition 'given a secondary student is chosen' restricts the denominator to the secondary column total: 150. Students who favor Math or Science among secondary students: 80 (Math) + 10 (Science) = 90. Probability = 90 / 150 = 3/5.

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35 **Answer: C** 6

Height $h = 4d$. The radius $r = d/2$. Volume of cone: $V = \frac{1}{3}\pi r^2 h = 72\pi \Rightarrow \frac{1}{3}(\frac{d}{2})^2(4d) = 72 \Rightarrow \frac{1}{3}(\frac{d^2}{4})(4d) = 72 \Rightarrow \frac{d^3}{3} = 72 \Rightarrow d^3 = 216$. Thus, $d = 6$.

36 **Answer: C** $\frac{1}{2}$

Subtract the first equation from the second: $(17x - 13y) - (11x - 7y) = 9 - 6 \Rightarrow 6x - 6y = 3 \Rightarrow 6(x - y) = 3 \Rightarrow x - y = \frac{3}{6} = \frac{1}{2}$.

37 **Answer: A** $h \leq \frac{14.5 - 1.4p - 3.5c}{2}$

Total spent cannot exceed $\$20$.
The cost of the items is $1.4p + 3.5c + 2h + 5.5$.
Inequality: $1.4p + 3.5c + 2h + 5.5 \leq 20$.
Subtract 5.5 from both sides: $1.4p + 3.5c + 2h \leq 14.5$.
Isolate $2h$: $2h \leq 14.5 - 1.4p - 3.5c$.
Divide by 2: $h \leq \frac{14.5 - 1.4p - 3.5c}{2}$.

38 **Answer: C** 25%

Let the initial price be 100. After a 20% discount, the price is 80. To return to 100, we must increase by 20. The percentage increase is $20/80 = 1/4 = 25\%$.

39 **Answer: C** 27

Sum of given percentages: $35 + 20 + 30 = 85\%$. Labor party makes up the remaining 15%. If $15\% = 27$ members, $1\% = 27/15 = 1.8$ members. Democrats (35%) minus conservatives (20%) is a 15% difference. A 15% difference equals exactly 27 members.

40 **Answer: B** -5

At $x = -1$, by carefully examining the grid points: $f(-1) = -1$, $g(-1) = -2$, and $h(-1) = -2$. Sum = $-1 + (-2) + (-2) = -5$.

41 **Answer: C** 3 days, 9 hours

Total walls = $18 \times 5.4 = 97.2$ walls. Total hours required = $97.2/1.2 = 81$ hours. We know 1 day = 24 hours. $81/24 = 3$ remainder 9. So 3 days and 9 hours.

42 **Answer: A** $320,000

Transport is 12.5%. $12.5\% = 200,000$. The total budget is $200,000/0.125 = 1,600,000$. Salaries represent 20%. 20% of 1,600,000 = $320,000$.

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43 **Answer: C**

See the answer key.

44 **Answer: A**

See the answer key.

45 **Answer: A**

See the answer key.

46 **Answer: C**

See the answer key.

47 **Answer: C** $\sqrt[8]{3}$

Rewrite applying exponent rules: $\frac{3^{1/4}}{3^{1/8}} = 3^{1/4-1/8} = 3^{2/8-1/8} = 3^{1/8} = \sqrt[8]{3}$.

48 **Answer: C** 2646

Percent increase = $(2520 - 2400)/2400 = 120/2400 = 0.05$ (5%). Next month increase will be 5% of 2520: $2520 \times 1.05 = 2646$.

49 **Answer: A** -12

Multiply by 3: $2(x + 2) - 3x > 15 \Rightarrow 2x + 4 - 3x > 15 \Rightarrow -x > 11 \Rightarrow x < -11$. The only option completely less than -11 is -12.

50 **Answer: B** 130

We convert square feet to square meters by dividing by $(3.28)^2$. $1\ \text{m}^2 = (3.28\ \text{ft})^2 = 10.7584\ \text{ft}^2$. $1400/10.7584 \approx 130.13$. So he needs approximately 130 square meters.

51 **Answer: 13**

Sum of x,y,z,t = $4 \times 11.5 = 46$. Sum of x,y,z,t,q = $5 \times 11.8 = 59$. Therefore, $q = 59 - 46 = 13$.

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52 **Answer: 7**

First, find k. $g(-2) = 3(-2)^2 + k(-2) - 8 = -4 \Rightarrow 3(4) - 2k - 8 = -4 \Rightarrow 12 - 2k - 8 = -4 \Rightarrow 4 - 2k = -4 \Rightarrow 2k = 8 \Rightarrow k = 4$. So $g(x) = 3x^2 + 4x - 8$. Now find $g(-3)$: $g(-3) = 3(-3)^2 + 4(-3) - 8 = 3(9) - 12 - 8 = 27 - 20 = 7$.

53 **Answer: 1**

By the Remainder Theorem, the remainder of dividing $P(x)$ by $x - 1$ is $P(1)$. $P(1) = 4(1)^3 - (1)^2 - 8(1) + 6 = 4 - 1 - 8 + 6 = 1$.

54 **Answer: 75.6**

Let C = kM. $1.5 = k(54) \Rightarrow k = 1.5/54$. Therefore, $2.1 = (1.5/54)M \Rightarrow M = 2.1 \times 54/1.5 = 75.6$g.

55 **Answer: 0.28**

$k(-2) = -2(-2) - 5 = 4 - 5 = -1$. Then $h(-1) = -(-1)^2 + 3(-1) - 2 = -1 - 3 - 2 = -6$.

56 **Answer: 1**

First, look up $q(0)$. From the table, when $x = 0$, $q(x) = -1$. Next, look up $p(-1)$. From the table, when $x = -1$, $p(x) = 1$. Thus, $p(q(0)) = 1$.

57 **Answer: 6**

$k(-2) = -2(-2) - 5 = 4 - 5 = -1$. Then $h(-1) = -(-1)^2 + 3(-1) - 2 = -1 - 3 - 2 = -6$.

58 **Answer: 32.5**

Groceries: 25% of 600 =150. Remaining amount = $450. Clothes$:30\%of450 = $135. Balance$ =450 - 135 = 315. Utensils = $120. Final balance for gaming accessories$ =315 - 120 = $195. Percentage of original$ =(195 / 600) \times 100 = 32.5\%$.

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